

Exercise (Security of ElGamal implies Decisional Diffie Hellman). *Show that ElGamal cryptosystem is (t, ε) -IND-CPA secure only if the underlying group $\mathbb{G}$ is Decisional Diffie Hellman group.*

Solution.

Hint: You need to convert DDH adversary to IND-CPA adversary. Can you see Diffie-Hellman tuples in IND-CPA game?